



CORPORATE CREDIT CARD MISUSE DETECTION USING MACHINE LEARNING AND ETL PIPELINE

CS6140 – FINAL PROJECT

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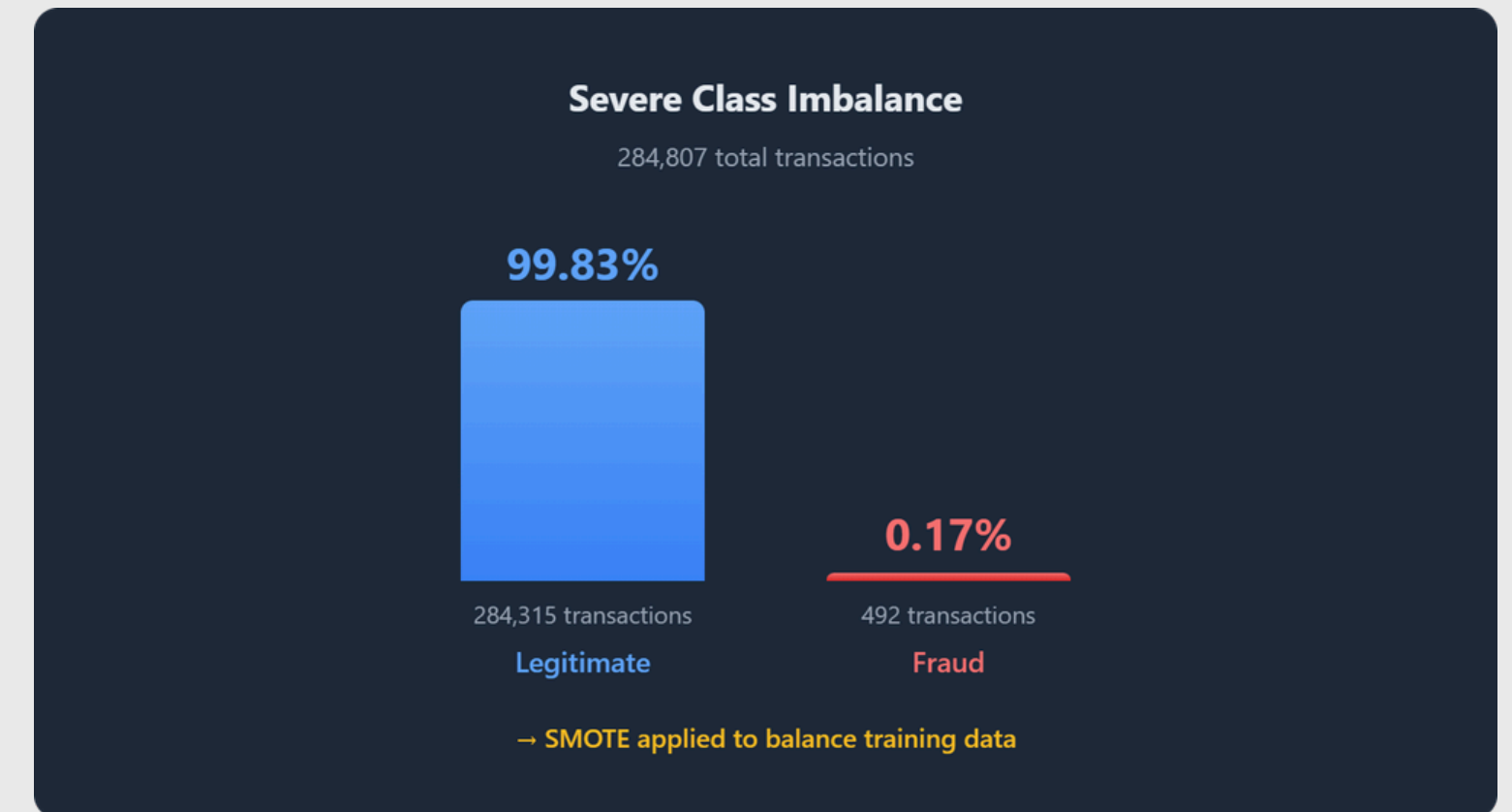


WHY THIS MATTERS

- 5% of revenue lost to fraud annually
- 284,807 transactions – only 0.17% are fraud
- Manual review? Not scalable.
- Arrow pointing to: Solution: ML-powered automated detection

ABOUT THE DATA

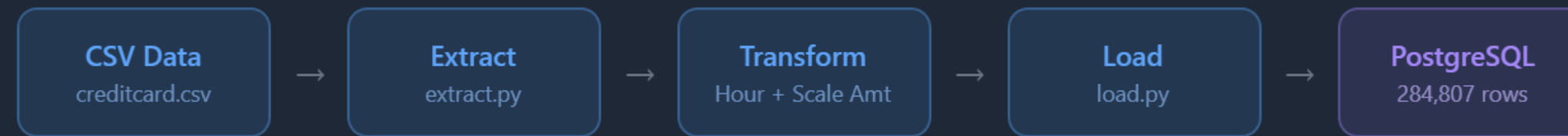
- Dataset: Credit Card Fraud Detection (ULB, Kaggle)
- 284,807 transactions
- 30 features: V1–V28 (PCA), Amount_Scaled, Hour
- Class imbalance visual: big blue bar (99.83% legit) vs tiny red bar (0.17% fraud)
- SMOTE applied to balance training data



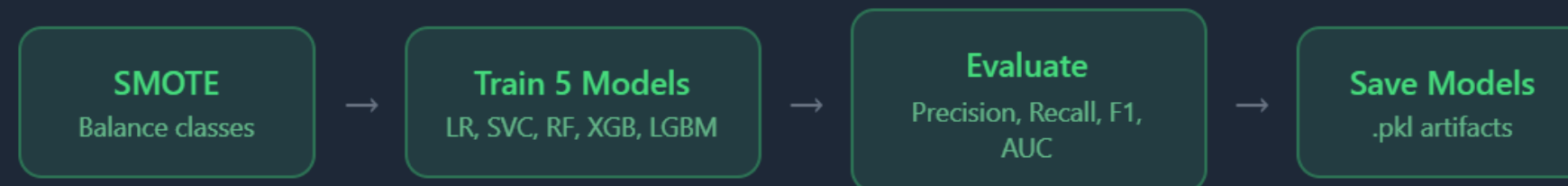
End-to-End Pipeline

End-to-End System Architecture

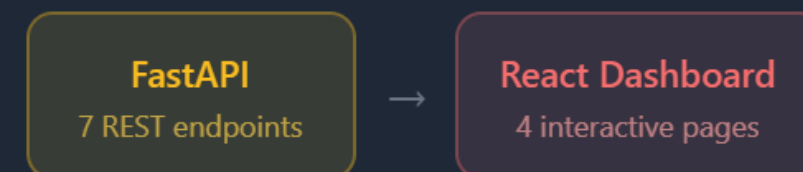
DATA INGESTION & ETL PIPELINE



MACHINE LEARNING TRAINING



SERVING & FRONTEND



■ ETL Pipeline ■ ML Training ■ API ■ Frontend ■ Database

Components to list:

- ETL Pipeline: Python scripts → PostgreSQL
- ML Training: 5 models with SMOTE
- Backend: FastAPI with 7 endpoints
- Frontend: React dashboard with 4 pages

Model Selection

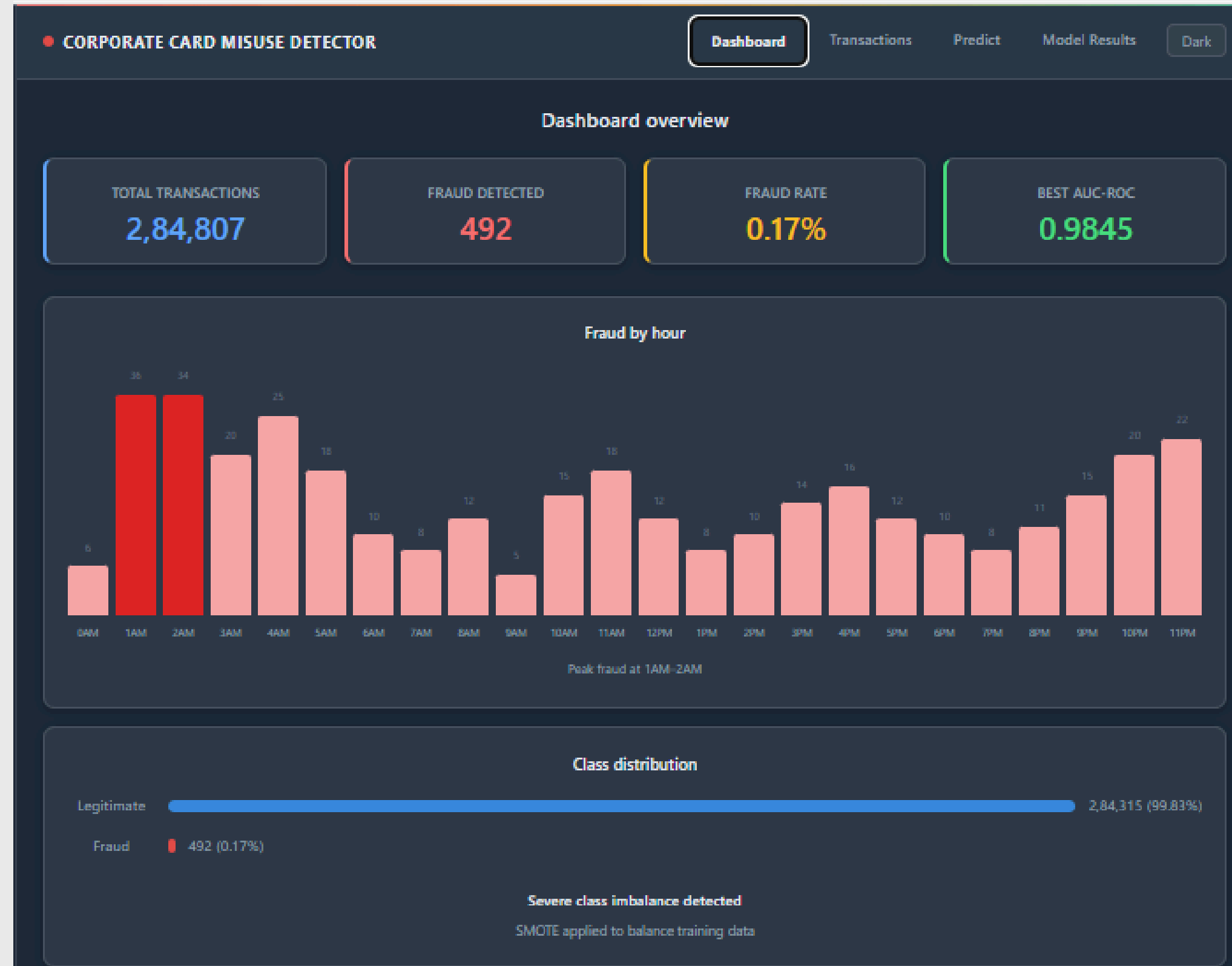
Model	Type	Why chosen
Logistic Regression	Baseline	Simple, interpretable
LinearSVC	Linear	Fast, margin-based
Random Forest	Ensemble	Handles imbalance naturally
XGBoost	Boosting	Sequential error correction
LightGBM	Boosting	Fast leaf-wise growth

RESULTS & ANALYSIS



Key insight: Random Forest wins best precision (0.86), F1 (0.84), and AUC (0.9845)

Output Screenshots



Dashboard

CORPORATE CARD MISUSE DETECTOR

Dashboard

Transactions

Predict

Model Results

Dark

All transactions

All

Fraud

Legitimate

Showing 1–15 of 2,84,807

ID	Amount	Hour	Status
T00001	\$227.44	1:00	Legitimate
T00002	\$29.91	1:00	Legitimate
T00003	\$31.03	1:00	Legitimate
T00004	\$31.73	1:00	Legitimate
T00005	\$22.23	1:00	Legitimate
T00006	\$34.92	1:00	Legitimate
T00007	\$14.33	1:00	Legitimate
T00008	\$34.25	1:00	Legitimate
T00009	\$15.05	1:00	Legitimate
T00010	\$34.81	1:00	Legitimate
T00011	\$30.20	1:00	Legitimate
T00012	\$7.18	1:00	Legitimate
T00013	\$25.73	1:00	Legitimate
T00014	\$56.63	1:00	Legitimate
T00015	\$24.36	1:00	Legitimate

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Transactions

CORPORATE CARD MISUSE DETECTOR

DashboardTransactionsPredictModel ResultsDark

Predict transaction

Select any transaction and click Predict — all 30 features are sent automatically for accurate prediction.

AllFraudLegitimateShowing 1–15 of 2,84,807

ID	Amount	Hour	Ground Truth	Action
T00001	\$227.44	1:00	Legitimate	Predict
T00002	\$29.91	1:00	Legitimate	Predict
T00003	\$31.03	1:00	Legitimate	Predicted
T00004	\$31.73	1:00	Legitimate	Predict
T00005	\$22.23	1:00	Legitimate	Predict
T00006	\$34.92	1:00	Legitimate	Predict
T00007	\$14.33			Predict
T00008	\$34.25			Predict
T00009	\$15.05			Predict
T00010	\$34.81			Predict
T00011	\$30.20			Predict
T00012	\$7.18			Predict
T00013	\$25.73			Predict
T00014	\$56.63	1:00	Legitimate	Predict
T00015	\$24.36	1:00	Legitimate	Predict

×

✓

Legitimate transaction

Fraud probability: 0.0%

Predicted: Legitimate

Ground Truth: Legitimate

✓ Correct

Model: Random Forest | All 30 features used

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Predict-legitimate case

CORPORATE CARD MISUSE DETECTOR

DashboardTransactionsPredictModel ResultsDark

Predict transaction

Select any transaction and click Predict — all 30 features are sent automatically for accurate prediction.

AllFraudLegitimateShowing 1–15 of 492

ID	Amount	Hour	Ground Truth	Action
T00001	\$11.73	1:00	Fraud	Predict
T00002	\$34.92	2:00	Fraud	Predict
T00003	\$34.92	2:00	Fraud	Predicted
T00004	\$34.92	2:00	Fraud	Predict
T00005	\$34.92	2:00	Fraud	Predict
T00006	\$34.92	2:00	Fraud	Predict
T00007	\$34.92			Predict
T00008	\$34.92			Predict
T00009	\$34.92			Predict
T00010	\$34.92			Predict
T00011	\$34.92			Predict
T00012	\$34.92			Predict
T00013	\$34.92			Predict
T00014	\$34.92	2:00	Fraud	Predict
T00015	\$34.92	2:00	Fraud	Predict

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Fraudulent transaction

Fraud probability: 100.0%

Predicted: Fraud

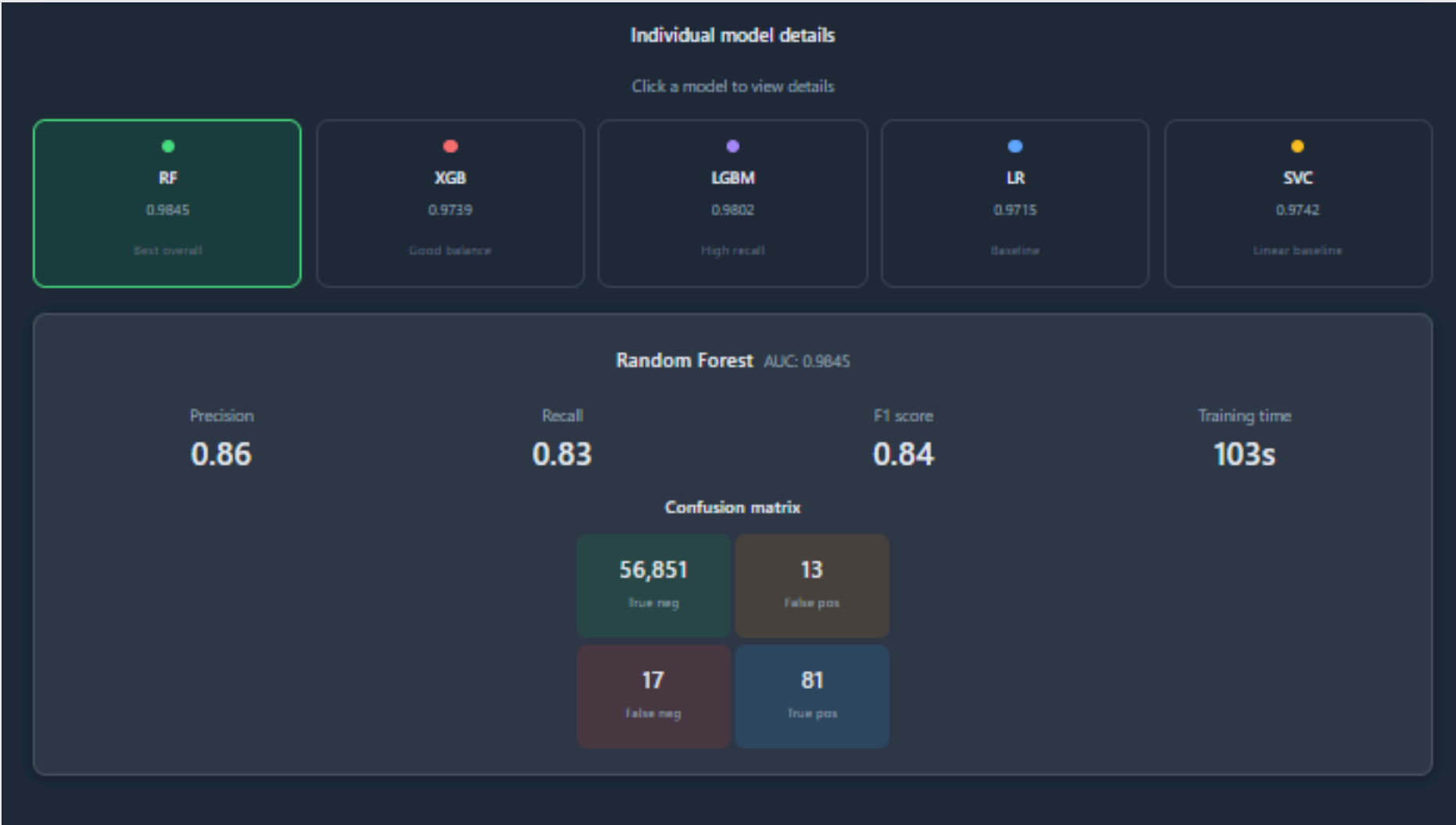
Ground Truth: Fraud

✓ Correct

Model: Random Forest | All 30 features used

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Predict- fraud case



Model Results

Limitations & Next Steps

Limitations:

- Dataset uses anonymized PCA features – can't interpret what V1–V28 represent
- Amount and Hour alone can't detect fraud – V features carry the signal
- No real-time streaming

Future work:

- Real-time streaming with Apache Kafka
- Airflow for scheduled pipeline orchestration
- Allowing user input



THANK YOU!

GitHub:

- github.com/pooja-malakappa-nuchchi/corp-card-fraud-pipeline
- <https://github.com/pooja-malakappa-nuchchi/corp-card-fraud-pipeline/tree/pooja>
- **Tech Stack: Python, PostgreSQL, FastAPI, React, scikit-learn, XGBoost, LightGBM**
- **"Questions? Post in the discussion thread!"**